THE STRUCTURE OF MATHEMATICAL REALITY

Phase II — Chapter 6

The Unreasonable Effectiveness Problem

Exercises

Self-Study Curriculum

1. Wigner gives the example of the use of complex numbers in quantum mechanics as "unreasonable." Hamming gives a different explanation for why complex numbers appear in physics (related to the structure of the equations). Read both arguments. Which do you find more convincing? Is there a synthesis?

2. In your own field, find an example of a mathematical tool that was adopted from another domain and turned out to work "unreasonably well." Describe what structural features of your system make the borrowed tool effective.

3. The compression argument says: a model works when the system has low complexity relative to its size. Can you think of a system in your experience where this is *not* true, where the system genuinely has high complexity relative to its size, and modeling it is correspondingly hard? What does this tell you about the boundaries of mathematical modeling?